Chapter-1

INTORDUCTION

1.1 Overview

With the rapid growth of two-wheeler usage, road accidents have become a major safety concern worldwide. Many accidents result in severe injuries or fatalities, often due to the lack of immediate medical assistance and failure to use proper safety gear like helmets. To address these issues, an IoT-based smart helmet system has been developed that integrates advanced technologies to improve rider safety.

The concept of the Internet of Things (IoT) enables devices to communicate and share data over the internet. By incorporating IoT into helmets, it becomes possible to monitor rider conditions, detect accidents in real time, and automatically send alerts to emergency contacts.

The smart helmet is equipped with sensors such as an accelerometer for accident detection, a GPS module for real-time location tracking, and a GSM module for sending alerts via messages. In case of an accident, the system detects sudden impact or abnormal motion and immediately transmits the rider’s location to predefined contacts or emergency services.

Additionally, some smart helmets include features like alcohol detection, helmet-wearing detection, and engine ignition control, ensuring that the rider follows safety regulations before starting the vehicle.

Overall, this system aims to reduce accident fatalities by enabling quick emergency response and promoting safer riding practices through intelligent monitoring and automation.

Another important aspect of the IoT-based smart helmet is its ability to provide real-time connectivity between the rider, vehicle, and emergency services. By continuously collecting and transmitting data, the system ensures that critical information such as location, speed, and impact intensity is readily available. This connectivity significantly reduces the response time during emergencies, which is often a crucial factor in saving lives.

The integration of wireless communication technologies plays a key role in the system’s effectiveness. Using modules like GSM or internet-based communication, the smart helmet can instantly notify family members, hospitals, or nearby authorities when an accident occurs. This automated alert system eliminates the dependency on bystanders, who may not always be present or able to provide timely help.

Moreover, the smart helmet can be enhanced with cloud-based data storage and monitoring. This allows the collection of historical data related to rider behavior, accident patterns, and route tracking. Such data can be analyzed to improve road safety measures, identify accident-prone areas, and support better traffic management systems

1.2 Motivation

Road accidents have become one of the most serious problems in modern transportation systems, especially for two-wheeler riders. Every year, thousands of people lose their lives due to delayed medical assistance after accidents. In many situations, accident victims remain unnoticed for a long time because nobody nearby informs emergency services immediately. Although helmets provide physical protection to riders, traditional helmets cannot communicate emergency information or track the rider's location during critical situations.

The motivation behind developing the IoT-Based Smart Helmet with Accident Detection and Location Tracking system is to enhance rider safety using modern technology. This project aims to reduce accident-related fatalities by automatically detecting accidents and sending emergency alerts with live location details to family members, hospitals, or rescue teams. The integration of IoT technology, GPS, GSM communication, and sensors provides a smart and intelligent safety solution for motorcycle riders.

Another major motivation for this project is the increasing use of smart devices and connected systems in daily life. By applying IoT technology to transportation safety, it becomes possible to create an automated emergency response system that works without human intervention. The project also encourages the use of embedded systems and real-time monitoring technologies in practical applications that benefit society.

This smart helmet system not only improves rider safety but also helps emergency services respond quickly, reducing the chances of serious injuries and deaths. Therefore, the project is motivated by the need to create a safer, smarter, and more reliable transportation environment for people around the world.

1.3 Problem statement

Road accidents involving two-wheeler riders are increasing rapidly due to over speeding, traffic congestion, careless driving, poor road conditions, and lack of immediate medical assistance. In many accident cases, victims are unable to contact family members or emergency services because of serious injuries or unconscious conditions. The delay in providing medical support during the "golden hour" after an accident often results in severe injuries or loss of life.

- Increasing number of motorcycle accidents every year.
- Delay in providing medical assistance to accident victims.
- Riders may become unconscious and unable to call for help.
- Traditional helmets provide only physical protection.
- Lack of automatic accident detection systems.

1.4 Objectives

- To design and develop an IoT-based smart helmet system.
- To detect accidents automatically using sensors.
- To track the rider's live location using GPS technology.
- To send emergency alert messages through GSM modules.
- To improve the safety of motorcycle riders.
- To reduce the delay in emergency medical assistance.
- To provide real-time monitoring during accidents.

1.5 Scope

1. Accident Detection

The smart helmet can automatically detect accidents using sensors such as accelerometers and vibration sensors. It identifies sudden impacts or abnormal movements and activates the emergency alert system immediately.

2. Real-Time Location Tracking

The system uses a GPS module to track the exact location of the rider during an accident. This helps rescue teams and family members locate the victim quickly.

3. Emergency Alert System

The GSM module sends emergency SMS alerts to predefined contacts along with the accident location details. This reduces delay in emergency response.

4. Rider Safety Improvement

The project enhances rider safety by combining traditional helmet protection with smart electronic monitoring and communication features.

5. Use in Multiple Applications

The smart helmet can be used by motorcycle riders, delivery agents, travelers, military personnel, and traffic safety departments for improved security and monitoring.

Chapter -2

LITERATURE SURVEY

Several researchers have worked on smart helmet systems to improve the safety of two-wheeler riders using Internet of Things technology. The main objective of these systems is to detect accidents, monitor rider safety, and provide quick emergency communication using sensors and wireless modules.

H.C. Impana, M. Hamsaveni, and H.T. Chethana proposed a smart helmet system for accident detection using IoT technology. Their work focused on using sensors, microcontrollers, and communication modules to improve rider safety and reduce accident response time. The study also compared different techniques such as image processing and sensor-based systems for helmet detection and accident monitoring. (EAI Endorsed Transactions)

Another research paper by Manjesh N and Prof. Sudarshan Raj introduced a smart helmet system using GPS and GSM technology. In this system, vibration sensors were used to detect crashes, while the GPS module tracked the rider's location. When an accident occurred, the GSM module automatically sent emergency messages to family members or ambulance services. (Research Publish)

Kabilan M., Monish S., and Dr. S. Siamala Devi developed an IoT-based accident detection smart helmet using sensors, Raspberry Pi, GPS, and emergency communication systems. Their system focused on improving real-time accident monitoring and emergency alert mechanisms for motorcycle riders. (IJARIIT)

Recent studies have also included additional safety features such as alcohol detection, helmet-wearing detection, and cloud-based monitoring systems. Researchers proposed systems using sensors like MPU6050, MQ-3 alcohol sensors, GSM, GPS, and IoT cloud platforms to improve accident prevention and emergency response. (IJPREMS)

From the literature survey, it is observed that smart helmet systems can significantly improve road safety by combining accident detection, location tracking, and emergency communication. However, challenges such as false accident detection, power management, compact design, and reliability still require improvement. The proposed IoT-based smart helmet system aims to provide a cost-effective, reliable, and efficient solution for rider safety and accident management.

Chapter-3

REQUIREMENT SPECIFICATION

1. Arduino Mega

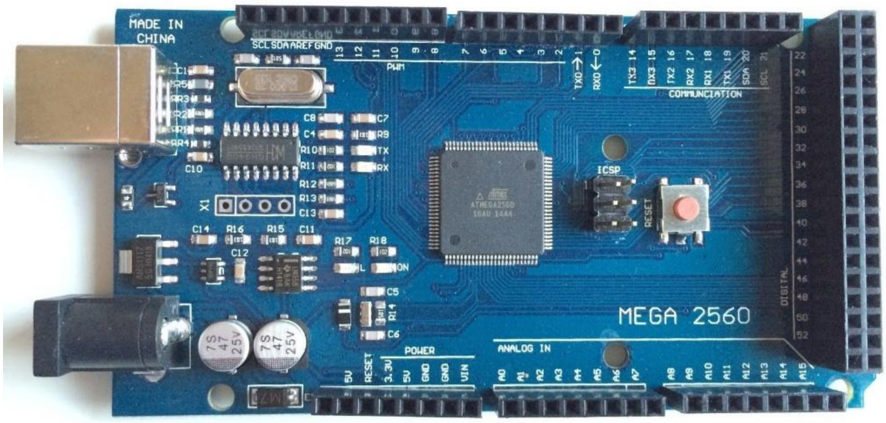

Fig 3.2 : Arduino Mega

Introduction

Arduino Mega is a microcontroller board based on the ATmega2560 controller. It is the main controlling unit of the smart helmet system and is used to connect and manage all input and output devices.

Features

- ATmega2560 microcontroller
- Large number of digital and analog pins
- Supports multiple serial communications
- Easy programming using Arduino IDE

Working Principle

The Arduino Mega receives signals from sensors such as MPU6050 and processes the data. Based on the sensor values, it controls the GSM module, GPS module, buzzer, and LED indicators.

Role in the Project

It acts as the brain of the smart helmet system. It continuously monitors accident conditions and controls the emergency alert process.

2. MPU6050

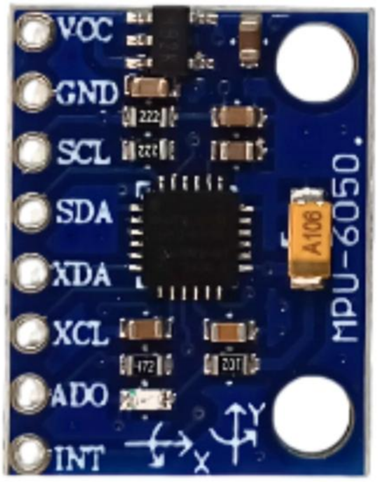

Fig 3.3 : MPU6050

Introduction

MPU6050 is a motion tracking sensor that combines an accelerometer and gyroscope in a single module. It is used for detecting movement, vibration, and tilt.

Features

- 3-axis accelerometer
- 3-axis gyroscope
- High sensitivity motion detection
- Compact and low power device

Working Principle

The sensor measures acceleration and angular movement of the helmet. Sudden impacts or abnormal tilt values are detected as accident conditions.

Role in the Project

It is used for accident detection. When a crash or sudden fall occurs, the sensor sends data to the Arduino Mega for emergency action.

3. NEO-6M GPS Module

Fig 3.4 : NEO-6M GPS module

Introduction

The NEO-6M GPS module is used to determine the geographical location of the rider using satellite signals.

Features

- Accurate latitude and longitude tracking
- Fast position detection
- Compact module design
- Supports real-time location monitoring

Working Principle

The GPS module receives signals from satellites and calculates the exact position of the rider. The location coordinates are sent to the Arduino controller.

Role in the Project

It provides the live accident location, which is included in the emergency alert message sent to emergency contacts.

4. SIM900A GSM Module

Fig 3.5 : SIM900A GSM module

Introduction

SIM900A is a GSM communication module used for sending SMS messages and making phone calls through a mobile network.

Features

- Supports SMS and voice call functions
- Compatible with Arduino boards
- Operates using SIM card network
- Reliable wireless communication

Working Principle

The GSM module receives commands from the Arduino Mega and sends SMS alerts or makes calls to predefined numbers during emergencies.

Role in the Project

It is responsible for emergency communication after accident detection and sends the rider's location details to family members or rescue teams.

5. 16x2 LCD Display

Fig 3.6 : 16x2 LCD display

Introduction

The 16x2 LCD display is used to show messages and system information.

Features

- Displays text and status messages
- Low power display module
- Easy interface with Arduino

Working Principle

The display receives commands from the Arduino Mega and shows information such as GPS status or alert messages.

Role in the Project

It helps users monitor the working condition and output status of the system.

6. Push Button

Fig 3.7 : Push Button

Introduction

The push button is a simple input device used for manual emergency activation.

Features

- Small and easy to use
- Quick response operation
- Low power consumption

Working Principle

When the button is pressed, it sends a signal to the Arduino Mega to activate emergency alerts manually.

Role in the Project

It helps the rider send emergency notifications even when an accident is not automatically detected.

7. LED Indicator

Fig 3.8 : LED Indicator

Introduction

LED is a light-emitting device used for visual indication in the smart helmet system.

Features

- Low power consumption
- Bright indication light
- Fast switching operation

Working Principle

The LED glows when it receives power from the Arduino controller, indicating system status or emergency conditions.

Role in the Project

It provides visual alerts and indicates the working condition of the system.

8. Buzzer

Fig 3.9 : Buzzer

Introduction

A buzzer is an electronic audio device used for generating alarm sounds.

Features

- Produces loud alert sound
- Compact and lightweight
- Easy interfacing with Arduino

Working Principle

The buzzer produces sound when it receives electrical signals from the Arduino Mega.

Role in the Project

It gives warning alarms during accident detection or emergency conditions.

9. Lithium-ion Battery

Fig 3.10 : Lithium-ion Battery

Introduction

The power supply unit is used to provide electrical energy to all the components of the IoT-based smart helmet system. A rechargeable lithium-ion battery is used because it is lightweight, portable, and capable of providing stable power for long durations.

Specifications

- Battery Type: Lithium-ion Rechargeable Battery
- Output Voltage: 12V DC
- Capacity: 2200mAh – 5000mAh
- Rechargeable: Yes

- Portable and lightweight
- Provides stable power supply to electronic components

Role in Providing Power to the System

The battery supplies power to the Arduino Mega, SIM900A GSM Module, NEO-6M GPS Module, buzzer, LED, and MPU6050. It ensures continuous operation of the smart helmet system during travel and emergency situations.

Chapter- 4

BLOCK DIAGRAM

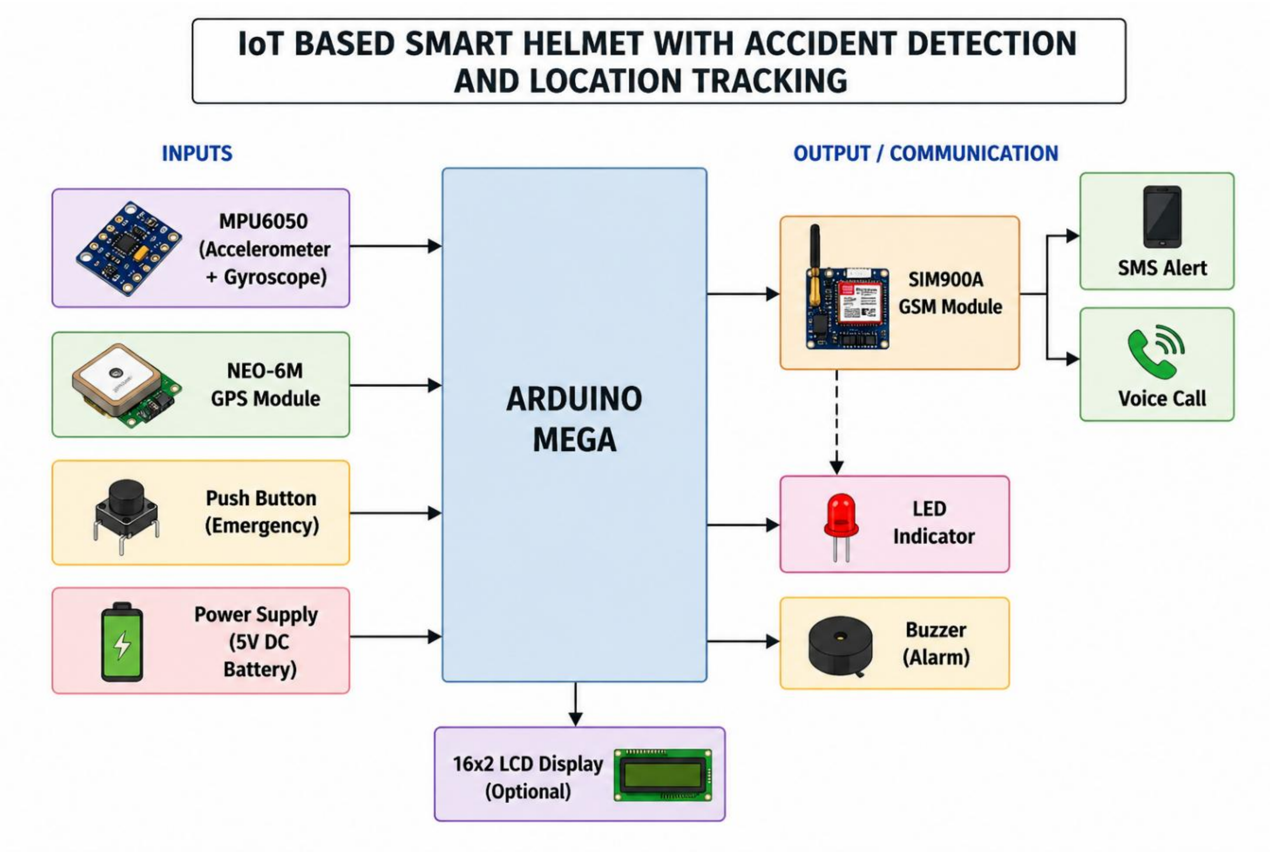

Fig 4.1 : Block diagram of IoT-based Smart Helmet with Accident Detection and Location Tracking.

Chapter-5

IMPLEMENTATION

1. Hardware Implementation

The hardware implementation of the IoT-Based Smart Helmet involves connecting various electronic components such as sensors, GPS module, GSM module, microcontroller, buzzer, and power supply inside the helmet. The accelerometer sensor is used to detect sudden impacts or abnormal movements caused by accidents. The GPS module collects the real-time location coordinates, while the GSM module sends emergency SMS alerts to predefined contacts.

The Arduino Uno or NodeMCU acts as the main controller of the system. All sensor data is processed by the controller, and necessary actions are performed automatically during emergency situations.

2. Software Implementation

The software implementation is carried out using Arduino IDE or Embedded C programming. The program continuously monitors sensor values and compares them with predefined threshold limits. When the sensor values exceed the threshold:

- The system identifies it as an accident.
- GPS coordinates are collected.
- GSM module sends an alert message with location details.
- Emergency buzzer gets activated.

The software also manages communication between all hardware modules efficiently.

3. Sensor Implementation

The accelerometer sensor (MPU6050) is mounted inside the helmet to monitor sudden acceleration and vibrations.

$$a = \frac{\Delta v}{\Delta t}$$

If the acceleration exceeds the safe limit, the system detects a collision or fall.

4. GPS and GSM Implementation

The GPS module continuously receives location data from satellites and provides latitude and longitude information. The GSM module uses a SIM card to send SMS alerts to emergency contacts.

Example SMS:

"Accident Detected! Rider needs help. Location: Latitude XX.XXXX, Longitude YY.YYYY"

This implementation ensures quick communication during emergencies.

5. Power Supply Implementation

The entire system is powered using a rechargeable battery connected to the helmet. Voltage regulators are used to provide stable power to the sensors and communication modules.

6. Testing and Verification

The system is tested under different conditions to verify:

- Accident detection accuracy
- GPS location accuracy
- SMS delivery speed
- Sensor response time
- Power consumption

Successful testing confirms that the smart helmet can effectively improve rider safety and emergency response.

Implementation Flow

1. Rider wears the smart helmet.
2. Sensors continuously monitor movement.
3. Accident or sudden impact occurs.
4. Microcontroller detects abnormal sensor values.
5. GPS module collects location details.
6. GSM module sends emergency alert message.
7. Rescue teams or family members receive notification.
8. Emergency assistance reaches the rider quickly.

Chapter-6

METHODOLOGY

1. **Power Supply Initialization**
   - The system is powered using a rechargeable lithium-ion battery.
   - The battery provides power to all electronic components of the smart helmet.
2. **System Startup**
   - After power is supplied, the Arduino Mega initializes all connected modules.
   - The controller checks the working status of the GSM, GPS, and sensor modules.
3. **Sensor Activation**
   - The MPU6050 starts monitoring the movement and orientation of the helmet.
   - It continuously measures acceleration, vibration, and tilt values.
4. **Real-Time Data Collection**
   - The sensor sends real-time motion data to the Arduino Mega.
   - The controller continuously processes the incoming data for accident analysis.
5. **Accident Detection**
   - If sudden impact or abnormal helmet movement exceeds the threshold value, the system detects an accident.
   - The Arduino immediately activates the emergency response process.
6. **GPS Location Tracking**
   - The NEO-6M GPS Module receives satellite signals to identify the rider's location.
   - It provides accurate latitude and longitude coordinates of the accident spot.
7. **Data Processing**
   - The Arduino Mega combines the accident information with GPS location data.
   - This information is prepared for emergency communication.
8. **SMS Alert Generation**
   - The SIM900A GSM Module receives commands from the controller.
   - It prepares and sends emergency SMS alerts to predefined mobile numbers.

9. Emergency Calling

- The GSM module can also make automatic phone calls during emergency situations.
- This helps family members or rescue teams receive immediate accident information.

10. Buzzer Activation

- The buzzer is activated after accident detection.
- It produces an alarm sound to indicate emergency conditions.

11. LED Indication

- LED indicators glow to show system status and emergency alerts.
- They help in visual identification of the system operation.

12. Manual Emergency Button

- A push button is provided for manual emergency activation.
- The rider can send alerts manually if immediate help is needed.

13. Continuous Monitoring

- The system continuously monitors the rider throughout the journey.
- All components work together in real time for better safety.

14. Emergency Response Support

- The location and alert message help rescue teams reach the accident spot quickly.
- This reduces delay in providing medical assistance.

15. System Reliability

- The smart helmet system is designed to operate efficiently with low power consumption.
- It provides reliable accident detection and communication support during emergencies.

Chapter-7

EXPERIMENTAL RESULTS & OUTPUT

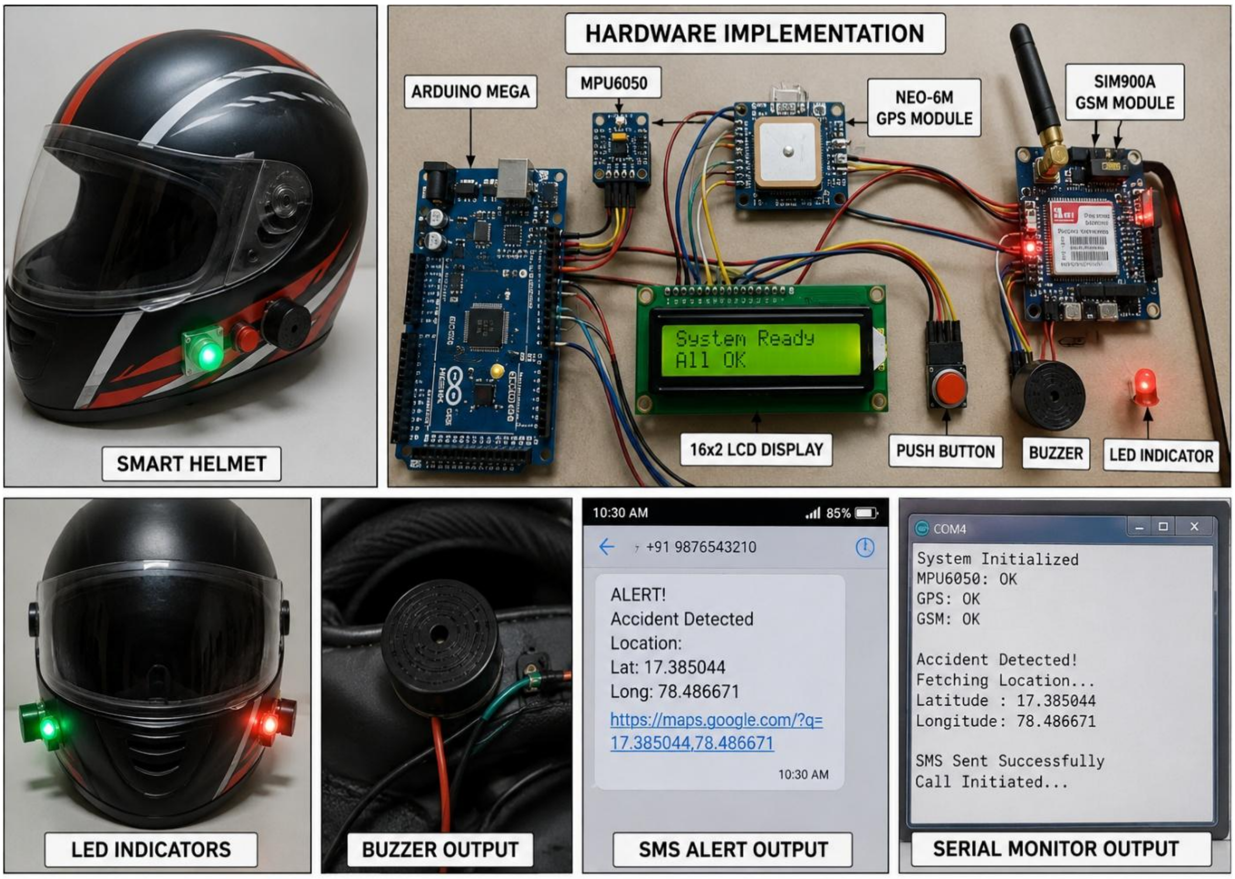

Fig 7.1 Circuit connection

RESULT

The IoT-based Smart Helmet with Accident Detection and Location Tracking system was successfully designed and implemented using Arduino Mega, MPU6050, NEO-6M GPS Module, and SIM900A GSM Module. The system successfully monitored helmet movement and detected accident conditions based on sudden impact and abnormal tilt values.

During testing, the MPU6050 sensor accurately detected crash movements and sent the data to the Arduino Mega for processing. Once an accident was detected, the system immediately activated the emergency response mechanism without manual intervention.

The GPS module successfully obtained the live latitude and longitude coordinates of the rider. The GSM module then sent emergency SMS alerts and initiated phone calls to predefined mobile numbers along with the accident location details.

The buzzer and LED indicators also worked properly by providing audio and visual alerts during emergency situations. The emergency push button feature successfully allowed manual alert activation whenever required.

The overall system operated effectively in real time and provided fast accident detection, location tracking, and emergency communication support. The project demonstrated that the proposed smart helmet system can improve rider safety, reduce emergency response time, and provide a reliable low-cost solution for accident management.

Chapter 8

APPLICATIONS

1. **Road Safety for Two-Wheeler Riders**
   The smart helmet improves the safety of motorcycle riders during travel.
   It helps in reducing injuries and fatalities during accidents.

2. **Accident Detection System**
   The system automatically detects road accidents using sensors.
   It provides quick emergency response during crash situations.

3. **Emergency Alert Communication**
   The helmet sends emergency SMS alerts and phone calls automatically.
   This helps family members and rescue teams receive accident information quickly.

4. **Real-Time Location Tracking**
   The GPS module provides the live location of the rider during emergencies.
   It helps ambulance services reach the accident spot faster.

5. **Smart Transportation Systems**
   The project can be used in modern intelligent transportation applications.
   It improves vehicle safety using Internet of Things technology.

6. **Delivery and Logistics Services**
   Delivery riders can use the smart helmet for additional safety during long-distance travel.
   It helps companies monitor emergency situations of riders.

7. **Traffic Safety Monitoring**
   Traffic departments can use the system for accident monitoring and emergency management.
   It supports faster response during road accidents.

8. **Personal Safety Applications**
   The helmet provides safety support for individual riders during night travel or remote area travel.
   Manual emergency alerts can be used during unsafe situations.

9. **Educational and Research Projects**
   The project can be used for engineering mini-projects and IoT research applications.
   It helps students learn sensor interfacing and communication technologies.

10. **Future Smart Helmet Development**
    The system can be upgraded with advanced features like cloud monitoring and health tracking.
    It provides a base for future intelligent helmet technologies.

ADVANTAGES

1. **Automatic Accident Detection**
   The system automatically detects accidents using sensors.
   This reduces the need for manual emergency reporting.

2. **Quick Emergency Response**
   Emergency alerts are sent immediately after accident detection.
   This helps in reducing delay in medical assistance.

3. **Real-Time Location Tracking**
   The GPS module provides accurate live location of the rider.
   It helps rescue teams reach the accident spot quickly.

4. **Improved Rider Safety**
   The smart helmet increases the safety of two-wheeler riders.
   It helps reduce the risk of severe injuries during accidents.

5. **Automatic SMS and Calling Feature**
   The GSM module sends SMS alerts and makes emergency calls automatically.
   This ensures fast communication during emergencies.

6. **Compact and Portable Design**
   The system is lightweight and can be easily fitted inside the helmet.
   It does not create discomfort for the rider.

7. **Low-Cost System**
   The project uses affordable electronic components.
   This makes it suitable for practical and large-scale use.

8. **Real-Time Monitoring**
   The system continuously monitors helmet movement and vibrations.
   It provides fast detection of accident conditions.

9. **Manual Emergency Support**
   The emergency push button allows manual alert activation.
   The rider can request help even without an accident.

10. **Easy to Operate**
    The smart helmet system is simple and user-friendly.
    It can be easily used by any motorcycle rider.

Chapter-9

Conclusion

The IoT-based Smart Helmet with Accident Detection and Location Tracking system was successfully developed to improve the safety of two-wheeler riders. The project combines Internet of Things technology with sensors and communication modules to provide automatic accident detection and emergency support.

The system effectively uses the MPU6050 to detect sudden impacts and abnormal helmet movements during accidents. The NEO-6M GPS Module accurately tracks the rider's location, while the SIM900A GSM Module sends emergency SMS alerts and phone calls to predefined contacts.

The project successfully achieved real-time monitoring, quick emergency communication, and reliable location tracking. Additional components such as the buzzer, LED indicators, and push button improved the functionality and usability of the system.

This smart helmet system helps reduce accident response time and increases the possibility of providing timely medical assistance to accident victims. The project also demonstrates that low-cost IoT technology can be effectively used for modern transportation safety applications.

In the future, the system can be enhanced by adding features such as alcohol detection, helmet-wearing detection, cloud connectivity, mobile application support, and health monitoring sensors.

Overall, the proposed smart helmet provides a reliable, efficient, and practical solution for improving road safety and emergency management for motorcycle riders.

BIBLIOGRAPHY

[1] Manjesh N and H.C. Impana, M. Hamsaveni, and H.T. Chethana, "Smart Helmet for Accident Detection and Notification using IoT," International Conference on IoT and Related Technologies, 2021.

[2] Prof. Sudarshan Raj, "Smart Helmet Using GSM and GPS Technology for Accident Detection and Reporting System," International Journal of Electrical and Electronics Research, Vol. 2, Issue 4, 2014.

[3] Kabilan M, Monish S, and Dr. S. Siamala Devi, "Accident Detection System Based on Internet of Things (IoT) Smart Helmet,"International Journal of Advance Research, Ideas and Innovations in Technology, 2020.

[4] "Arduino Mega 2560 Datasheet," Arduino Official Website

[5] "SIM900A GSM Module Technical Specifications," SIMCom Official Website

[6] "NEO-6M GPS Module User Guide," u-blox Official Website

[7] "MPU6050 Accelerometer and Gyroscope Sensor Datasheet," TDK InvenSense Official Website

[8] Internet of Things: A Hands-On Approach by Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015.

[9] Embedded Systems: Introduction to ARM Cortex-M Microcontrollers by Jonathan W. Valvano, CreateSpace Publications, 2016.

[10] Research articles and online technical references related to IoT-based accident detection systems, smart helmets, GPS tracking, and GSM communication modules.